

Murat Yessenov, Ph.D.

Curriculum Vitae



CURRENT AFFILIATION

Harvard University
John A. Paulson School of Engineering
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CONTACT INFORMATION

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EDUCATION

Ph.D. in Optics
2019 – 2022

The College of Optics and Photonics (CREOL)
University of Central Florida, Orlando, FL, USA
PhD committee: Ayman Abouraddy (PI), G. Atia, B. A. Saleh and D. Christodoulides
Dissertation title: “Physics and applications of space-time wave packets”

M.Sc. in Optics
2017 – 2019

The College of Optics and Photonics (CREOL)
University of Central Florida, Orlando, FL, USA

B.Sc. in Physics
2013 – 2017

Nazarbayev University, Astana, Kazakhstan

- Magna cum Laude Diploma
- Minor in Mathematics

RESEARCH EXPERIENCE

Feb ‘25 - Present

Post-doctoral Fellow at Harvard John A. Paulson School of Engineering and Applied Sciences, Harvard University, Cambridge, MA

PI: Federico Capasso

- Design and fabrication of high-NA (>0.8) metalens covering the whole visible range for applications in metrology, in collaboration with KLA Corporation.
- Design and fabrication of high-uniformity, high-damage threshold meta-optics for optical neutral atom traps with $>10,000$ traps, in collaboration with Prof. Misha Lukin’s group and QuEra Company.
- Fully programmable self-imaging platform based on Montgomery effect with applications in multi-plane imaging and optical atom traps.

May ‘22 – Feb ‘25

Post-doctoral Research Associate at The College of Optics and Photonics (CREOL), University of Central Florida, Orlando, FL, USA

PI: Ayman F. Abouraddy

- Miniature spectrometer based on Bragg gratings; particle acceleration with structured light sources; spatio-temporally structured single-photon fields, and spatio-temporally structured plasmon-polariton waves.
- Supervising 3 graduate students actively involving research metrics and goals, maintaining organization of the group progress, and guided in their research projects.
- Leads collaborative projects in conjunction with research groups at Harvard University, Stanford University, Tsukuba University, and other institutions.

July ‘17 – May ‘22

Graduate Research Assistant at The College of Optics and Photonics (CREOL), University of Central Florida, Orlando, FL, USA

PI: Ayman F. Abouraddy

- Worked on newly developed spatio-temporally shaped light waves.

- Developed theoretical models for free space propagation and nonlinear interaction of finite-energy spatio-temporally structured light fields. Created 20+ comprehensive optical simulations for modeling, design, DAQ and data analysis that continue to be used by students in the group.
- Published over 30 papers, with 12 as first author; presented at over 15 conferences
- Contributed to grant proposal for MURI, ONR and Keck Foundation, and research articles in high-impact journals (Physical Review Letters, Nature Photonics, Nature Communications, Optica, etc.)

VOLUNTEERING EXPERIENCE

2023 - present	Co-founder, Advisory Board member of Kazakhstani Association of PhDs in North America (KAPNA) A hub for networking, collaboration and support among PhD professionals from Kazakhstan currently pursuing a career in North America with 140+ members. Organized over 30 in-person and online events, leading the LinkedIn page.
2022 – 2023	Mentor at Nazarbayev University Alumni Mentorship Program. Providing mentorship for undergraduate and graduate students in overseas graduate program applications and guiding through their first steps into academic path.
2021 – 2022	Treasurer for IEEE Orlando Photonics Society Orlando chapter of IEEE professional society
2015 – 2017	Co-organizer of STEM camps among schools in Astana, Kazakhstan Co-organized over 10 intense 5-day education camps in physics, math, or English language. 3000+ total participants from 4 secondary schools in Astana, Kazakhstan.

AWARDS AND HONORS

February 2022	CREOL Student of the Year
October 2021	Boris P. Stoicheff Memorial Scholarship
May 2021	The Laser Technology, Engineering and Applications Scholarship by SPIE
September 2020	Emil Wolf Outstanding Student Paper Competition
2012 – 2013	Medalist in National and International Physics Olympiads, Kazakhstan

PROFESSIONAL SKILLS

Technical	Free-space optics alignment, pulse shaping, beam shaping, pulse characterization (FROG, autocorrelator), Mach-Zehnder interferometer, pump-probe measurements, fiber coupling. Accustomed to operating and performing maintenance of OSA, Oscilloscope, Lasers, Detector, Polishing tools.
Software	Languages/Software: CST Studio, Origin, MATLAB, Mathematica, C/C++, Python Graphical design: Photoshop, Blender, and Inkscape. EM simulations: BPM, FDTD, FDFD

PUBLICATIONS

RESEARCH ARTICLES

1. **Yessenov, M.**, Sacchi, L., Palmieri, A., Hall, L. A., Abouraddy, A. F., & Capasso, F. (2026). Observation of the spatially structured Montgomery effect in free space. *Optica*, 13(2), 195-202.
2. Hall, L. A., **Yessenov, M.**, Schepler, K. L., & Abouraddy, A. F. (2025). Universality and non-differentiability: A new perspective on angular dispersion in optics. *APL Photonics*, 10(12).
3. Ichiji, N., Kikuchi, H., **Yessenov, M.**, Schepler, K. L., Abouraddy, A. F., & Kubo, A. (2025). Observation of space-time surface plasmon polaritons. *Nature Communications*, 16(1), 10697.
4. Yaraghi, S., Mhibik, O., **Yessenov, M.**, Miller, J. K., Parto, M., Johnson, E. G., ... & Divliansky, I. (2025). Resonance-free Fabry-Pérot cavity via unrestricted orbital-angular-momentum ladder-up. *Nature Communications*, 16(1), 10362.
5. **Yessenov, M.**, Dorrah, A. H., Guo, C., Hall, L. A., Park, J. S., Free, J., ... & Abouraddy, A. F. (2025). Ultrafast space-time optical merons in momentum-energy space. *Nature Communications*, 16(1), 8592.
6. **Yessenov, M.**, & Abouraddy, A. F. (2025). Optical spatiotemporal Fourier synthesis: tutorial. *Journal of the Optical Society of America A*, 42(9), 1295-1315.
7. Hall, L. A., **Yessenov, M.**, Romer, M. A., & Abouraddy, T. B. A. F. (2025). Long-distance axial spectral encoding using space-time wave packets. *Optics Letters*, 50(15), 4698-4701.
8. Abouraddy, A. F., **Yessenov, M.**, Divliansky, I., & Watnik, A. T. (2025). New frontiers in spatiotemporally structured light. *Optics and Photonics News*, 36(7), 38-45.
9. Su, X., Zou, K., Wang, Y., **Yessenov, M.**, Zhou, H., Song, H., ... & Willner, A. E. (2025). Space-time wave packets in multimode optical fibers with controlled dynamic motions and tunable group velocities. *Nature Communications*, 16(1), 2027.
10. Ichiji, N., **Yessenov, M.**, Schepler, K. L., Abouraddy, A. F., & Kubo, A. (2024). Exciting space-time surface plasmon polaritons by irradiating a nanoslit structure. *JOSA A*, 41(3), 396-405.
11. **Yessenov, M.**, Romer, M., Ichiji, N., & Abouraddy, A. F. (2024). Experimental realization of Lorentz boosts of space-time wave packets. *Phys. Rev. A*, 109(1), 013509.
12. Mhibik, O., **Yessenov, M.**, Glebov, L., Abouraddy, A. F., & Divliansky, I. (2023). Compact dual-band spectral analysis via multiplexed rotated chirped volume Bragg gratings. *Opt. Lett.*, 48(19), 5137-5140.
13. Wan, Z., **Yessenov, M.**, & Padgett, M. J. (2023). The propagation speed of optical speckle. *Scientific Reports*, 13(1), 9071.
14. **Yessenov, M.**, & Abouraddy, A. F. (2023). Relativistic transformations of quasi-monochromatic paraxial optical beams. *Phys. Rev. A*, 107(4), 042221.
15. **Yessenov, M.**, Mhibik, O., Mach, L., Hayward, T. M., Menon, R., Glebov, L., ... & Abouraddy, A. F. (2022). Ultra-compact synthesis of space-time wave packets. *Opt. Lett.*, 48(10), 2500-2503.
16. Mhibik, O., **Yessenov, M.**, Mach, L., Glebov, L., Abouraddy, A. F., & Divliansky, I. (2023). Rotated chirped volume Bragg gratings for compact spectral analysis. *Opt. Lett.*, 48(5), 1180-1183.
17. Ichiji, N., Kikuchi, H., **Yessenov, M.**, Schepler, K. L., Abouraddy, A. F., & Kubo, A. (2023). Observation of ultrabroadband striped space-time surface plasmon polaritons. *ACS Photon.*, 10(2), 374-382.
18. Shen, Y., Zhan, Q., Wright, L. G., Christodoulides, D. N., Wise, F. W., Willner, A. E., ... & Forbes, A. (2023). Roadmap on spatiotemporal light fields. *J. of Opt.*, 25(9), 093001.
19. Minoofar, A., Zou, K., Pang, K., Song, H., Karpov, M., **Yessenov, M.**, ... & Willner, A. E. (2022). Generation of OAM-carrying space-time wave packets with time-dependent beam radii using a coherent combination of multiple LG modes on multiple frequencies. *Optics Express*, 30(25), 45267-45278.
20. Zou, K., Su, X., **Yessenov, M.**, Pang, K., Karapetyan, N., Karpov, M., ... & Willner, A. E. (2022). Tunability of space-time wave packet carrying tunable and dynamically changing OAM value. *Opt. Lett.*, 47(21), 5751-5754.
21. Pang, K., Zou, K., Zhao, Z., Song, H., Zhou, Y., Karpov, M., **Yessenov, M.** ... & Willner, A. E. (2022). Experimental demonstration of dynamic spatiotemporal structured beams that simultaneously exhibit two orbital angular momenta by combining multiple frequency lines, each carrying multiple Laguerre-Gaussian modes. *Opt. Lett.*, 47(16), 4044-4047.
22. **Yessenov, M.**, Chen, Z., Lavery, M. P., & Abouraddy, A. F. (2022). Vector space-time wave packets. *Opt. Lett.*, 47, 4131-4134.
23. Pang, K., Zou, K., Song, H., Karpov, M., **Yessenov, M.**, Zhao, Z., ... & Willner, A. E. (2022). Synthesis of near-diffraction-free orbital-angular-momentum space-time wave packets having a controllable group velocity using a frequency comb. *Opt. Express*, 30(10), 16712-16724.

24. **Yessenov, M.**, Hall, L. A., Schepler, K. L., & Abouraddy, A. F. (2022). Space-time wave packets. *Adv. Opt. Photon.* 14(3), 455-570.
25. **Yessenov, M.**, Faryadras, S., Benis, S., Hagan, D. J., Van Stryland, E. W., & Abouraddy, A. F. (2021). Refraction of space-time wave packets in a dispersive medium. *Opt. Lett.* 47 (7), 1630-1633.
26. **Yessenov, M.**, Free, J., Chen, Z., Johnson, E. G., Lavery, M. P., Alonso, M. A., & Abouraddy, A. F. (2021). Space-time wave packets localized in all dimensions. *Nat. Commun.* 13 (1), 1-13
27. Diouf, M., Harling, M., **Yessenov, M.**, Hall, L. A., Abouraddy, A. F., & Toussaint, K. C. (2021). Space-time vector light sheets. *Opt. Exp.*, 29(23), 37225-37233.
28. Hall, L. A., **Yessenov, M.**, and Abouraddy, A. F. (2021). Arbitrarily accelerating space-time wave packets, *Opt. Lett.*, 47 (7), 1630-1633.
29. **Yessenov, M.**, Bhaduri, B., & Abouraddy, A. F. (2021). Refraction of space-time wave packets: I. Theoretical principles. *J. Opt. Soc. Am. A* 38(10) 1409-1422.
30. Allende Motz A. A., **Yessenov, M.**, Bhaduri, B., and Abouraddy, A. F. (2021). Refraction of space-time wave packets: II. Experiments at normal incidence. *J. Opt. Soc. Am. A* 38(10) 1450-1461
31. **Yessenov, M.**, Allende Motz A. A., Bhaduri, B., and Abouraddy, A. F. (2021). Refraction of space-time wave packets: III. Experiments at oblique incidence. *J. Opt. Soc. Am. A* 38(10) 1462-1470
32. Hall, L. A., **Yessenov, M.**, Ponomarenko, S. A., & Abouraddy, A. F. (2021). The space-time Talbot effect. *APL Photonics* 6 (5), 056105.
33. **Yessenov, M.**, Hall, L. A., & Abouraddy, A. F. (2021). Engineering the optical vacuum: Arbitrary magnitude, sign, and order of dispersion in free space using space-time wave packets. *ACS Photon.*, 8, 8, 2274–2284
34. Allende Motz A. A., **Yessenov, M.**, and Abouraddy, A. F. (2021). Isochronous space-time wave packets, *Opt. Lett.* 46 (10), 2260-2263.
35. Hall, L. A., **Yessenov, M.**, and Abouraddy, A. F. (2021). Space–time wave packets violate the universal relationship between angular dispersion and pulse-front tilt, *Opt. Lett.* 46, 1672-1675.
36. Allende Motz A. A., **Yessenov, M.**, and Abouraddy, A. F. (2021). Axial spectral encoding of space-time wave packets. *Physical Review Applied*, 15(2), 024067.
37. **Yessenov, M.**, Hall, L. A., Ponomarenko, S. A., & Abouraddy, A. F. (2020). Veiled talbot effect. *Physical Review Letters*, 125(24), 243901.
38. **Yessenov, M.**, & Abouraddy, A. (2020). Accelerating and decelerating space-time optical wave packets in free space, *Physical Review Letters.*, 125, 233901.
39. **Yessenov, M.**, Bhaduri, B., Delfyett, P. J., & Abouraddy, A. F. (2020). Free-space optical delay line using space-time wave packets. *Nature Communications*, 11(1), 1-10.
40. Shiri, A., **Yessenov, M.**, Webster, S. et al. Hybrid guided space-time optical modes in unpatterned films. *Nature Communications* 11, 6273 (2020).
41. Bhaduri, B., **Yessenov, M.**, & Abouraddy, A. F. (2020). Anomalous refraction of optical spacetime wave packets. *Nature Photonics*, 14, 416–421 (2020)
42. Shiri, A., **Yessenov, M.**, Aravindakshan, R., & Abouraddy, A. F. (2020). Omni-resonant space–time wave packets. *Optics Letters*, 45(7), 1774-1777.
43. **Yessenov, M.**, Ru, Q., Schepler, K. L., Meem, M., Menon, R., Vodopyanov, K. L., & Abouraddy, A. F. (2020). Mid-infrared diffraction-free space-time wave packets. *OSA Continuum*, 3(3), 420-429.
44. Schepler, K., **Yessenov, M.**, Zhiyenbayev, Y., & Abouraddy, A. F. (2020). Space–Time Surface Plasmon Polaritons: A New Propagation-Invariant Surface Wave Packet. *ACS Photonics*.
45. **Yessenov, M.**, & Abouraddy, A. F. (2019). Changing the speed of optical coherence in free space. *Optics letters*, 44(21), 5125-5128.
46. **Yessenov, M.**, Bhaduri, B., Kondakci, H. E., Meem, M., Menon, R., & Abouraddy, A. F. (2019). Non-diffracting broadband incoherent space–time fields. *Optica*, 6(5), 598-607.
47. **Yessenov, M.**, Bhaduri, B., Kondakci, H. E., & Abouraddy, A. F. (2019). Weaving the rainbow: Space-time optical wave packets. *Optics and Photonics News*, 30(5), 34-41.
48. **Yessenov, M.**, Mach, L., Bhaduri, B., Mardani, D., Kondakci, H. E., Atia, G. K., ... & Abouraddy, A. F. (2019). What is the maximum differential group delay achievable by a space-time wave packet in free space? *Optics express*, 27(9), 12443-12457.
49. Bhaduri, B., **Yessenov, M.**, Reyes, D., Pena, J., Meem, M., Fairchild, S. R., ... & Abouraddy, A. F. (2019). Broadband space-time wave packets propagating 70 m. *Optics letters*, 44(8), 2073-2076.
50. **Yessenov, M.**, Bhaduri, B., Kondakci, H. E., & Abouraddy, A. F. (2019). Classification of propagation-invariant space-time wave packets in free space: Theory and experiments. *Physical Review A*, 99(2), 023856.
51. Bhaduri, B., **Yessenov, M.**, & Abouraddy, A. F. (2019). Space–time wave packets that travel in optical materials at the speed of light in vacuum. *Optica*, 6(2), 139-146.

52. Bhaduri, B., **Yessenov, M.**, & Abouraddy, A. F. (2018). Meters-long propagation of diffraction-free space-time light-sheets. *Optics express*, 26(16), 20111-20121.
53. Kondakci, H. E., **Yessenov, M.**, Meem, M., Reyes, D., Thul, D., Fairchild, S. R., ... & Abouraddy, A. F. (2018). Synthesizing broadband propagation-invariant space-time wave packets using transmissive phase plates. *Optics Express*, 26(10), 13628-13638.

PATENTS

1. Divliansky, I., Abouraddy, A., Glebov, L., Mhibik, O. and **Yessenov, M.**, University of Central Florida Research Foundation, Inc., 2024. SYSTEMS AND METHODS USING A PAIR OF ROTATED VOLUME BRAGG GRATINGS FOR SPECTRAL PHASE MODULATION. U.S. Patent Application 18/532,922.
2. Divliansky, I., Abouraddy, A., Glebov, L., Mhibik, O. and **Yessenov, M.**, University of Central Florida Research Foundation, Inc., 2024. SPECTROMETER WITH ROTATED VOLUME BRAGG GRATING. U.S. Patent Application 18/532,698.
3. Divliansky, I., Abouraddy, A., Glebov, L., Mhibik, O. and **Yessenov, M.**, University of Central Florida Research Foundation, Inc., 2024. SYSTEMS AND METHODS USING A PAIR OF ROTATED VOLUME BRAGG GRATINGS FOR LASER AMPLIFICATION. U.S. Patent Application 18/532,983.

CONFERENCE PROCEEDINGS

1. Mhibik, O., **Yessenov, M.**, Abourady, A., & Divliansky, I. (2025, August). Rotated Chirped Volume Bragg Gratings for Sensing and Space-Time Applications. In 2025 IEEE Research and Applications of Photonics in Defense Conference (RAPID) (pp. 1-2). IEEE.
2. Yaraghi, S., Mhibik, O., **Yessenov, M.**, Abourady, A., & Divliansky, I. (2025, August). Resonance-Free Fabry-Perot Cavity via Orbital Angular Momentum Ladder-Up. In 2025 IEEE Research and Applications of Photonics in Defense Conference (RAPID) (pp. 1-2). IEEE.
3. **Yessenov, M.**, Dorrah, A. H., Guo, C., Hall, L. A., Park, J. S., Free, J., ... & Abouraddy, A. F. (2025, May). Space-time optical merons in free space. In CLEO: Science and Innovations (p. SS148_3). Optica Publishing Group.
4. **Yessenov, M.**, Mhibik, O., Mach, L., Hayward, T. M., Menon, R., Glebov, L., ... & Abouraddy, A. F. (2025, May). Ultra-compact synthesis of spatiotemporal optical fields. In CLEO: Science and Innovations (p. SS148_1). Optica Publishing Group.
5. Ren, H., Lee, J., **Yessenov, M.**, Liu, Y. G., Abouraddy, A. F., Christodoulides, D. N., & Khajavikhan, M. (2025, May). On-chip generation of incoherent non-diffracting space-time optical fields. In CLEO: Fundamental Science (p. PD100_5). Optica Publishing Group.
6. **Yessenov, M.**, Mhibik, O., Hayward, T. M., Menon, R., Divliansky, I., & Abouraddy, A. F. (2024, September). Compact synthesis setup for space-time wave packets. In Laser Science (pp. JTu5A-32). Optica Publishing Group.
7. Yaraghi, S., Mhibik, O., **Yessenov, M.**, Divliansky, I., & Glebov, L. (2024, May). Design and fabrication of rotated chirped Bragg gratings. In *Practical Holography XXXVIII: Displays, Materials, and Applications* (p. PC1291004). SPIE.
8. Mhibik, O., **Yessenov, M.**, Mach, L., Glebov, L., Abouraddy, A. F., & Divliansky, I. (2024, March). Rotated chirped Bragg gratings: design and applications. In *Components and Packaging for Laser Systems X* (p. PC1286605). SPIE.
9. **Yessenov, M.**, Mhibik, O., Mach, L., Glebov, L. B., Abouraddy, A. F., & Divliansky, I. B. (2023, June). Compact spectrometer based on a rotated chirped volume Bragg grating. In *Next-Generation Spectroscopic Technologies XV* (Vol. 12516, p. 125160H). SPIE.
10. **Yessenov, M.**, Mhibik, O., Mach, L., Hayward, T. M., Menon, R., Glebov, L., ... & Abouraddy, A. F. (2023, May). Demonstration of ultra-compact setup for synthesis of space-time wave packets. In *CLEO: Fundamental Science* (pp. FTu4D-6). Optica Publishing Group.
11. **Yessenov, M.**, Mhibik, O., Mach, L., Glebov, L., Abouraddy, A. F., & Divliansky, I. (2023, March). Compact ultra-fast pulse modulation device based on rotated chirped volume Bragg gratings. In *Frontiers in Ultrafast Optics: Biomedical, Scientific, and Industrial Applications XXIII* (p. PC124110V). SPIE.

12. **Yessenov, M.**, Mhibik, O., Mach, L., Haywar, T. M., Menon, R., Glebov, L., ... & Abouraddy, A. F. (2023, March). Ultra-compact and robust setup for synthesis of space-time wave packets. In *Complex Light and Optical Forces XVII* (p. PC124360G). SPIE.
13. **Yessenov, M.**, Free, J., Chen, Z., Johnson, E. G., Lavery, M. P., & Abouraddy, A. F. (2022, December). 3D propagation-invariant space-time wave packets. In *Holography, Diffractive Optics, and Applications XII* (Vol. 12318, p. 123180N). SPIE.
14. **Yessenov, M.**, Free, J., Chen, Z., Johnson, E. G., Lavery, M. P., Alonso, M. A., & Abouraddy, A. F. (2022, September). Demonstration of propagation-invariant 3D space-time wave packets. In *European Physical Journal Web of Conferences* (Vol. 266, p. 08017).
15. **Yessenov, M.**, Chen, Z., Free, J., Johnson, E., Lavery, M., & Abouraddy, A. (2022). Demonstration of space-time wave packets localized in all dimensions. *Bulletin of the American Physical Society*.
16. Schepler, K., **Yessenov, M.**, & Abouraddy, A. (2022). Subwavelength propagation-invariant surface plasmon polaritons. *Bulletin of the American Physical Society*.
17. **Yessenov, M.**, Free, J., Chen, Z., Johnson, E. G., Lavery, M. P., Alonso, M. A., & Abouraddy, A. F. (2022, May). Diffraction-free Space-Time Wave Packets Localized in All Dimensions. In *CLEO: Applications and Technology* (pp. JTh4Q-3). Optica Publishing Group.
18. **Yessenov, M.**, Hall, L. A., & Abouraddy, A. F. (2022, May). Realizing Arbitrary Dispersion Profiles in Free Space Using Space-time Wave Packets. In *CLEO: Science and Innovations* (pp. STh2E-2). Optica Publishing Group.
19. Ichiji, N., **Yessenov, M.**, Schepler, K. L., Abouraddy, A. F., & Kubo, A. (2022, March). Propagation analysis of space-time surface plasmon polariton wave packet excited by light irradiation on nanobeam structure. In *Ultrafast Phenomena and Nanophotonics XXVI* (Vol. 11999, pp. 14-20). SPIE.
20. **Yessenov, M.**, Chen, Z., Free, J., Johnson, E. G., Lavery, M. P., & Abouraddy, A. F. (2022, March). Demonstration of 3D space-time wave packets. In *Complex Light and Optical Forces XVI* (p. PC1201703). SPIE.
21. **Yessenov, M.**, & Abouraddy, A. F. (2021). Space-time wave packets: A new class of spatio-temporally structured light waves. In 2021 IEEE Photonics Society Summer Topicals Meeting Series (SUM) (pp. 1-2). IEEE.
22. Hall, L. A., **Yessenov, M.**, Ponomarenko, S. A., & Abouraddy, A. F. (2021). Demonstration of the space-time Talbot effect. In *CLEO: QELS_Fundamental Science* (pp. FF2H-3). Optical Society of America.
23. Hall, L. A., **Yessenov, M.**, Ponomarenko, S. A., & Abouraddy, A. F. (2021). Demonstration of the space-time Talbot effect. In *CLEO: QELS_Fundamental Science* (pp. FF2H-3). Optical Society of America.
24. **Yessenov, M.**, Bhaduri, B., Kondakci, E. H., Meem, M., Menon, R., & Abouraddy, A. F. (2021). Demonstration of broadband diffraction-free incoherent space-time light sheets. In *Image Sensing Technologies: Materials, Devices, Systems, and Applications VIII* (Vol. 11723, p. 117230K), SPIE.
25. **Yessenov, M.**, Hall, L., Bhaduri, B., Meem, M., Schepler, K. L., Menon, R., ... & Abouraddy, A. (2021). Extended propagation of sub-mm incoherent space-time fields over 110 m distance. In *Laser Technology for Defense and Security XVI* (Vol. 11724, p. 1172409), SPIE.
26. Schepler, K. L., **Yessenov, M.**, Zhiyenbayev, Y., & Abouraddy, A. F. (2021, April). Guideless optical sensing using space-time surface plasmon polaritons. In *Fiber Optic Sensors and Applications XVII* (Vol. 11739, p. 1173904), SPIE.
27. **Yessenov, M.**, Hall, L. A., Ponomarenko, S. A., & Abouraddy, A. F. (2020, September). Demonstration of a Veiled Space-Time Talbot Effect. In *Frontiers in Optics* (pp. FW4B-1). Optical Society of America.
28. Motz, A. M. A., **Yessenov, M.**, & Abouraddy, A. F. (2020, September). Programmable Axial Spectral Evolution of Space-Time Wavepackets. In *Frontiers in Optics* (pp. FTu6C-3). Optical Society of America.
29. Schepler, K. L., Zhiyenbayev, Y., **Yessenov, M.**, & Abouraddy, A. F. (2020, September). Sub-wavelength Space-Time Surface Plasmon Polaritons on Unpatterned Surfaces. In *Frontiers in Optics* (pp. FM4E-4). Optical Society of America.
30. Shiri, A., **Yessenov, M.**, Aravindakshan, R., & Abouraddy, A. F. (2020, September). Omni-Resonant Space-Time Wave Packets with Programmable Fabry-Pérot-Cavity Spectra. In *Frontiers in Optics* (pp. FTh5B-4). Optical Society of America.
31. **Yessenov, M.**, Hall, L., Bhaduri, B., Reyes, D., Pena, J., Meem, M., ... & Abouraddy, A. F. (2020, September). Demonstration of Diffraction-free Incoherent Space-time Fields Propagating 110 Meters. In *Frontiers in Optics* (pp. FW1A-6). Optical Society of America.
32. **Yessenov, M.**, Bhaduri, B., & Abouraddy, A. F. (2020, September). Anomalous Refraction of Space-time Wave Packets. In *Frontiers in Optics* (pp. FW4B-2). Optical Society of America.
33. **Yessenov, M.**, & Abouraddy, A. F. (2020, September). Tuning the Coherence Group Velocity in Free Space. In *Frontiers in Optics* (pp. FW1A-1). Optical Society of America.

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